INDUSTRIAL TRAINING FINAL REPORT

POLITEKNIK BANTING SELANGOR

PERSIARAN ILMU,
JALAN SULTAN ABDUL SAMAD,
42700, BANTING
SELANGOR DARUL EHSAN

PREPARED BY:

Name :Muhammad Afiq Bin Roslan

Matric No :2015886518

Programme :EE112

Training Period :17Apr-9Jun

Supervisor

: Mohd.Azhar Bin Mohd.Khairi

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By:

MUHAMMAD AFIQ BIN ROSLAN

2015886518

Fulfilled the training at:

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An industrial training report submitted for partial fulfilment of the requirement for the Diploma in Electrical engineering

June 2017

ACKNOWLEDGEMENT

First of all,I would like to express my highest gratitude to Allah S.W.T for His guidance,bless and for giving me the strength to perform my responsibilities as a trainee and complete this industrial training report within the stipulated time.

My grateful thanks goes to the Puan Salina binti Ahmad for giving me the opportunity to carry out my industrial training in this agency for 8 weeks period. A big contribution and hard worked fro staff during practical period is very great indeed.

The special thank goes to my helpful supervisor,Ahmad Aminuddin bin Basar.The supervision and support that he gave truly help the progression and smoothness of the internship programme.Not forget, great appreciation go to the rest of Mohd.Azhar bin Mohd.Khairi, staff that help me from time to time during the project.The whole program really brought us to appreciate the true value of friendship and respect of each other.

Great deals appreciated go to the contribution of Department of Unit Pembangunan dan Penyelenggaraan.My grateful thanks to Puan Siti Azura Ramlan as my lecturer that take responsibility for this internship training because he really hard worked to make sure that my internship place was accepted. Not forget to my other lecturer that always give support and motivation to undergo this internship.

Lastly, to my family and friends that also give me strength and guide to continue my journey as a trainee and complete my internship. Thank you to all of you that involve as direct or indirect to make sure this programme smooth and clear.

ABSTRACT

This report is explain about my activities that I have done during industrial training for 8 week in Politeknik Banting Selangor. Politeknik Banting Selangor (PBS) were the places for student to get a diploma or degree certificate. It also explain about development and maintenance unit that which I doing industrial training with monitored by supervisor to make our job more safety and can be done smoothly.

Inside this report it will explain more detail about background of the company, history of the company, overview of industrial training at Politeknik Banting Selangor, and my activities during industrial training, problems encountered with comment, and how to solve the problems with my supervisor. Beside that,every safety, component and equipment will be explain to make more clearly know how to use and to be more aware on surrounding.

This report were making with my full hardworking and struggle to finish it because I'm narrowness of time between my part time job and doing report. Hopefully, every information that will be explain can satisfying the reader at once can give benefits to myself and the organization itself. Other than that, it can give me the direction and strong to face every problems in my future life at work environment.

TABLE OF CONTENT

CHAPTER 1: INTRODUCTION

1.1 Introduction of firm

Organization names : Politeknik Banting Selangor

Registered Address : POLITEKNIK BANTING SELANGOR,
PERSIARAN ILMU,
JALAN SULTAN ABDUL SAMAD,
42700, BANTING
SELANGOR DARUL EHSAN

Telephone No :03-31201625

Fax No :03-31201706

E-mail :admin@polibanting.edu.my

Website :http://www.polibanting.edu.my

Types Of Practice :Development and maintenance unit.

1.2 Company Background

1.2.1 HISTORY

Politeknik Banting Selangor (PBS) is a polytechnic to 24 of the 32 polytechnics in Malaysia. PBS has been operational at its temporary campus in 2007 by taking the first cohort in January 2008. The operation of the polytechnic session has started in full on 6 September 2012 in its permanent campus by offering two programs, namely Diploma in Aircraft Maintenance Engineering and Diploma in Mechanical Engineering. The first intake of students on campus are still in session in December 2012 with a total enrollment of 165 students.

PBS has always given priority to the sharing of knowledge and industry relationships in which teachers are encouraged to participate in industrial projects, research and development. This effort is open to the exchange of ideas and expertise with institutions and industry to produce graduates skilled, competent, ready to work and have the nature for marketability. Hence, the PBS intends to develop the network and collaborative relationships with the industry on an ongoing basis. In addition, students will be able to equip themselves for more industry savvy, resilient with the right mind set towards achieving high-income nation's workforce. This initiative will continue to be a priority to strengthen the resilience and competitiveness of students.

In realizing this mission, PBS has appointed 10 members of the Advisory Board of the caliber to join together in achieving the organization's direction and advise on the management and improve the quality of graduates PBS. Party PBS feel proud and lucky and thank the contributions made by members of the Advisory Board in reinforcing and improving the programs offered at the university in line with the rapid technological advancements.

1.2.2 MISION

To produce a holistic human capital to ensure the marketability of graduates high

1.2.3 VISION

Being highly respected technical training institutions by 2020

1.2.4 QUALITY BASE

PBS was determined to develop the excellence of students to ensure that students acquire the marketability and career readiness as well as meeting the needs of customers through the process of continuous improvements

1.2.5 Company list of product or service

Product

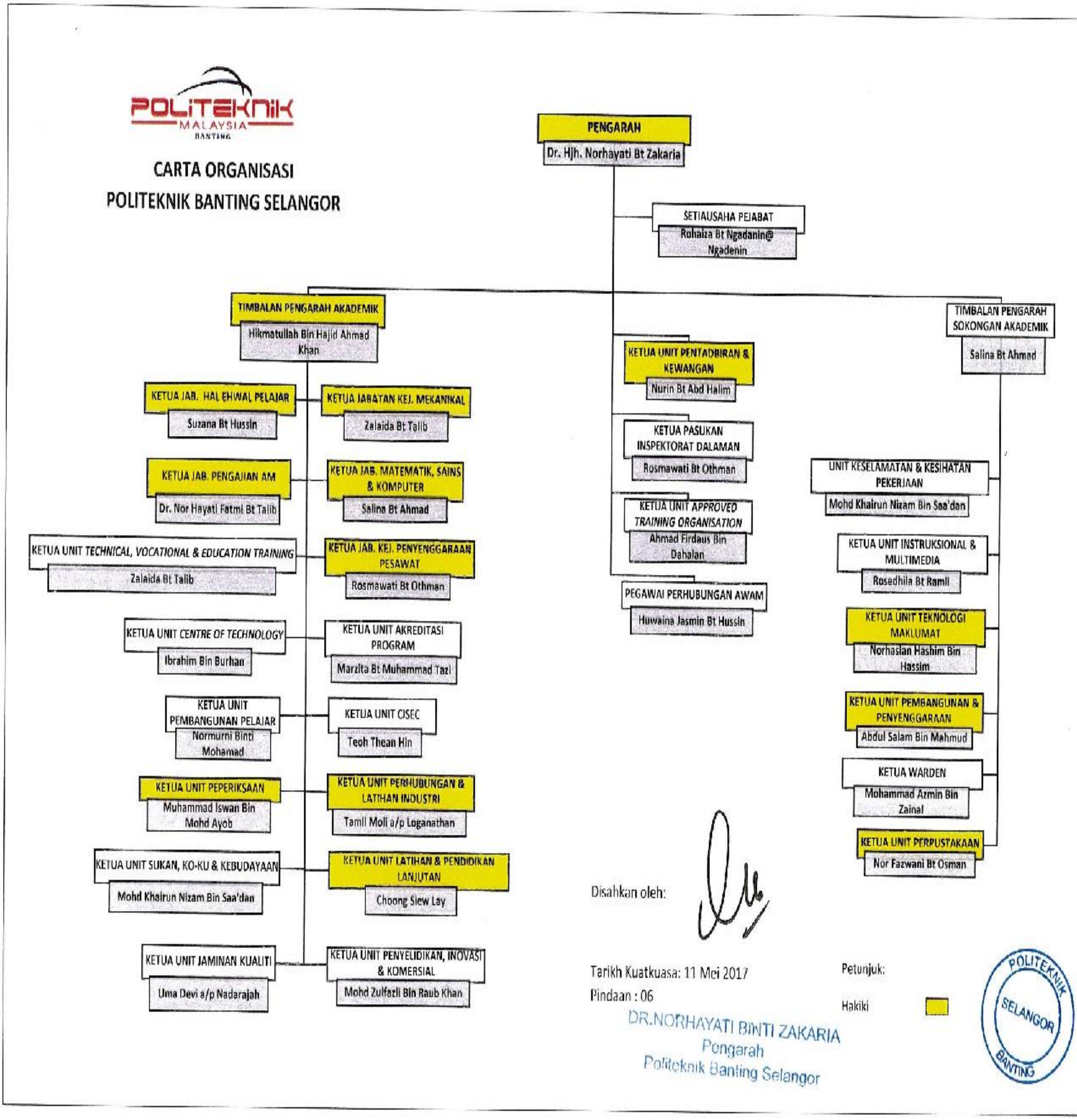
-Student,it will train the student to gain more knowledge.

Service

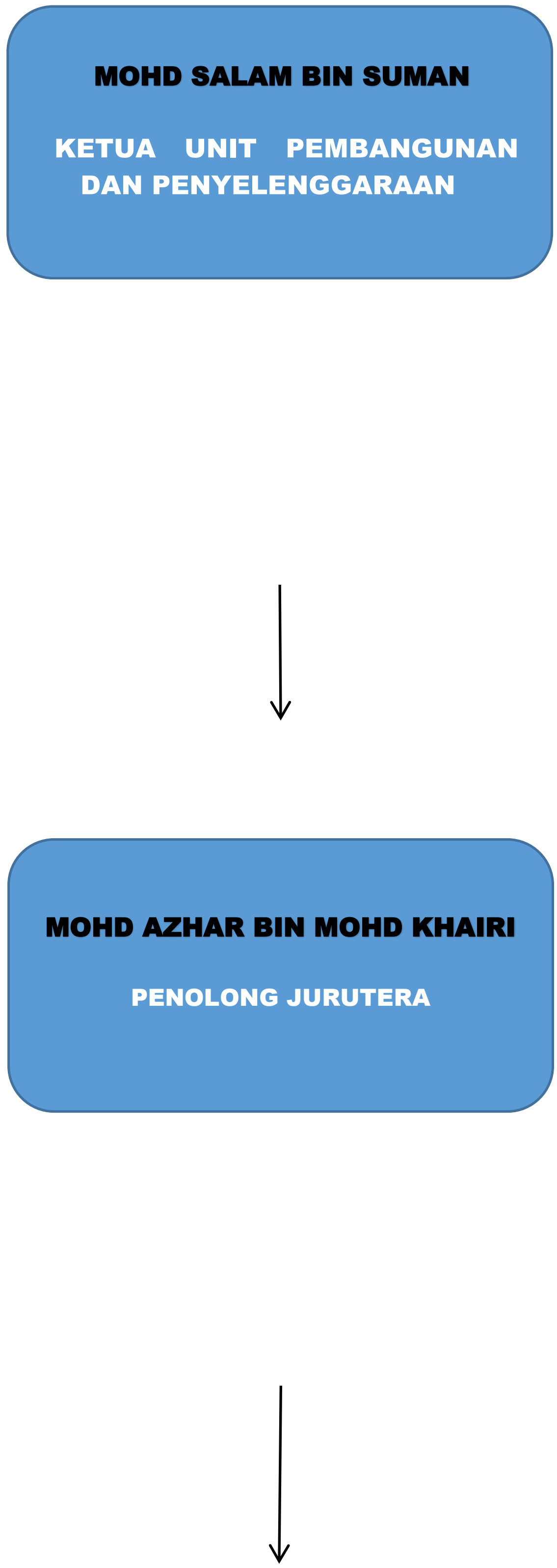
-Teaching and learning.

1.3 <u>Organization chart.</u>

1.3.1 Organization chart Politeknik Banting Selangor.



1.3.2 Organization chart Development and Maintenance Unit.



AHMAD AMINUDDIN BIN BASAR

JURUTEKNIK

CHAPTER 2: JOB OVERVIEW

Industrial Training module is a main component in the learning curriculum for Universiti Teknologi Mara (UITM). Industrial training is one of the compulsory courses for every UITM students.Every students bounds to be involved in industrial training for one whole semester in order for him or her to get his or her diploma certificate.

2.1 Purpose and objective of training.

2.1.1 Purpose

The purpose of industrial training is to produce graduates who are ready and capable to face their profession academically or non-academically with a high professionalism appearance. Other than that, the industrial training exposes the students about the real situation of the working class citizen. The industrial training also helps in developing social skills in the students.

2.1.2 Industrial Training Objective

To expose the students to the real life working experience and expanding the knowledge in their specific field.

i. Students will further learn about their real life profession. They will also learn what they need to do in order to finish their works. This will prepare the students so that they will easily fit in and fulfill the demands of their profession after they finish their course.

To make use of the theory and learned in the UITM.

ii. Students are only exposed to the basic theory and needed in the fields of their own. These theory and were mainly according to the books. With the industrial training in place, students will experience real life situation in the field. This will make the students use their knowledge in order to get their works done.

To produce trustworthy workers with high responsibility and able to cooperate with other staffs.

i. All tasks given by the supervisor must be completed with a sense of trustworthy and full responsibility. This attitude very important to ensure all business entrusted to students were carried out flawlessly. Also, it train the students to be honest not only to themselves but others as well.

Enchance student’s confidence at end of the training.

ii. When the industrial training students are exposed to a variety of problems and had to face it. With the experience learned through industrial training, the students will be more confidence both in learning and working. High spirit and skills to overcome the problems faced certainly create a strong confidence in the students.

Learn to interact with superior officers.

i. In this training, students get the opportunity to interact with upper management such as engineer,director and consultant and others. With this opportunity , student will be able to associate and discuss with them in a closer way. This opportunity will not come without the implication of industrial training. This opportunity should be used well by the students to learn more.

Increase the interest and curiosity.

i. There will be many things that students will face during the period of the industrial training. These events will enhance the curiosity in the students. Other than that,the real life exposure given to the students will also increase the interest of the students towards their fields of studies.

2.2 Scope Of Work

Unit pembangunan dan penyenggaraan (UPP) is a unit that manages the development,

Upgrading of buildings and infrastructure as well as maintenance of buildings and infrastructure.UPP is responsible for ensuring that all development and physical facilities to work in parallel with the development of teaching and learning (P&P) to ensure that the learning process can proceed smothly.

Development and maintenance unit prioritize maintenance to guarantee an asset which can provide satisfactory returns.Assets must be maintained more effectively and professionally to obtain best value for money.The maintenance of an effective requires an approach which is more strategic and comprehensive

Assets will not provide a satisfactory return to it if maintenance is neglected or maintenance will be performed only when there is serious damage or complain from users in the asset does not reach the optimum/sickly newly repaired and this can interfere with productivity.

Maintenance should be emphasized especially important assets involved in the learning process of polytechnics Malaysia disturbed and avoid long term loss.

Objective

Ensure that all buildings and infrastructure facilities in good condition and safe to support the process of semi-professional training and running perfectly smoothly.

Function

To coordinate the requirements and procurement services for upgrading works,renovation,repair work and maintenance of the physical facilities of general polytechnic.

2.3 Job responsibilities

-Take a action to fix a damage occured.

-Compare the cost for fix damaged occured to get the low budget.

-To make sure all condition around Politeknik always safe.

-Always make a routine inspection.

-Dealing with contractor for better repair work.

OVERVIEW OF TRAINING

WEEK 1

17 April 2017	• Self-report. • Checking the trip in lab.
18 April 2017	• Make a inspection routine. • Filing the outgoing and incoming mail.
19 April 2017	• Make a inspection routine. • Sending a form to the financial unit. • Make a monitoring of L.O manufacture.
20 April 2017	• Make a inspection routine. • Troubleshoot the hanger door with supervisor.

21 April 2017	• Wiring the plug.

Table 1.0:Week 1 overview industrial training

WEEK 2

25 April 2017	• Wiring the switch. • Maintenance the air condition in library. • Collaborative activities.

26 April 2017	• Repair a whiteboard. • Install a locking door at Bilik Kuliah. • Make a inspection routine.
27 April 2017	• Medical leave.
28 April 2017	• Medical leave.

Table 1.1:Week 2 overview industrial training

WEEK 3

2 May 2017	• Troubleshoot with contractor air condition at hanger,gym,makmal sains and infront JKM.
3 May 2017	• Troubleshoot the panel board for air condition at hanger. • Make a inspection routine.
4 May 2017	• Filing • Make a inspection routine. • Changing lamp at admin.
5 May 2017	• Make a inspection routine. • Learning from supervisor how to troubleshoot the trip at Feeder Pillar.

Table 1.2:Week 3 overview industrial training

WEEK 4

8 May 2017	● Troubleshoot the hanger door with contractor.
9 May 2017	● Troubleshoot the lamp at Bilik Kuliah. ● Fix the chair at canteen.
10 May 2017	● Public holiday (Wesak day).

11 May 2017	• Filing • Mail check routine daily inspection hostel. • Fix the chair at canteen.
12 May 2017	• Changing the battery bus.

Table 1.3:Week 4 overview industrial training

WEEK 5

15 May 2017	• Mailing the letter of air conditioning problems to other company. • Sending the invoice of local order to finance department to pay the contractor.
16 May 2017	• Changing the inner tube of motorcycles. • Fix the trip air condition and time at punch card machine.
17 May 2017	• Drilling the wall to make a place holder for extinguisher. • Checking the compressor.
18 May 2017	• Fix the barrier gate. • Troubleshoot the trip at feeder pillar.
19 May 2017	• Fix the water sprinkler infront admin. • Take a photo the damaged ceiling at around building. • Estimate the distance of damaged ceiling to know the area damage.

Table 1.4:Week 5 overview industrial training

WEEK 6

22 May 2017	• Monitor contractor repair the leak roof. • Make a inspection routine.
23 May 2017	• Send the mail quotation to Hazrin electrical company. • Fix the leaking pipe. • checking the compressor.
24 May 2017	• wiring the plug socket. • Drilling the table to make hole for vise table.
25 May 2017	• Staff meetings for photography and association. • Checking the barbed wire at the below of the wall.

26 May 2017	• Learning the location address at distribution board. • Repeal the partition.

Table 1.5:Week 6 overview industrial training

WEEK 7

29 May 2017	• Make a inspection routine. • Troubleshoot the trip at socket pantry in JKM. • Fix the flush in toilet library. • Learning the “simple Self-Diagnosis by malfunction Code” for air condition. • Sending the mail quotation to Azam Motor Workshop Sdn.Bhd.

30 May 2017	• Troubleshoot the trip at Kooperasi. • Make a inspection routine. • troubleshoot the fountain nozzle at pond.
31 May 2017	• Sending the mail quotation to finance department. • cleaning the pond.
1 June 2017	• Record the letter of local order before sending to finance department. • Troubleshoot the socket at Dewan. • Filing the outgoing and incoming mail.
2 June 2017	• Checking the water leaking at Dewan and hanger. • Fix the damaged light switch. • Changing the broken socket with new socket.

Table 1.6:Week 7 overview industrial training

WEEK 8

5 June 2017	• Learn how to write quotation mail. • checking the VRV with contractor Daikin air condition. • Checking the barbed wire. • Sending quotation mail to ninety-nine company.
6 June 2017	• Sending quotation mail to finance department. • Fixing the compressor
7 June 2017	• Sending quotation mail to finance department. • Changing the inner tube for motorcycle.
8 June 2017	• Fill in the letter of local order before sending to finance department. • Make a new telephone line connector.
9 June 2017	• Make a inspection routine. • End of industrial training.

Table 1.7:Week 8 overview industrial training

CHAPTER 3: TECHNICAL REPORT

3.1 Introduction

This chapter is explain about the whole weekend activities that have been done during practical. Based on activities that have been record on log book with briefly will be explain in this chapter was task from my supervisor to be done. Every activities or task that have been done must record at log book. That's why this part will be explain more detail about activities that have been done during industrial training at Politeknik Banting for 2 month.

All activities will be explain for week by week , so that can concluded the works in a week ,other than that it can easier the process of review. Work day in a week not always same each week because the company have a public holiday or holiday that will be announcement from company.

During undergoing the industrial training,I have been exposed with more new things. Besides that,I had been introduced with the rules and company ethics where I’m doing the industrial training. During my industrial training I have been places to the development and maintenance unit department that more to repair and maintenance works. My supervisor is mechanical engineer but in their he must learn electrical and civil engineering to do the job as technician.

Each work task had been scheduled by maintenance engineer for easier our job maintenance. After that every week we do prevention maintenance that’s mean routine check around the Politeknik Banting by followed the scheduled. In undergo industrial training, I have gave time for 8 hour per day,start from 8.00 a.m until 5.00 p.m every day for Monday until Friday. That’s was my weekly summary when I’m doing the industrial training.

3.2 FEEDER PILLARS

Figure 1.1: Feeder pillar in Politeknik Banting

Definition

A feeder pillar provides local isolation to your electrical distribution equipment, protecting both the cabling and the transformer from faults. Feeder pillars allow simple and local maintenance to your equipment, reducing site downtime and reducing overall maintenance costs. We can offer a comprehensive range of feeder pillars which come fully assembled to meet your specification, allowing easy on site installation.

Feeder pillar is weatherproof for outdoor installation,it free standing,cable connected or transformer mounted. Voltage for the feeder pillar is 415V that use 3 phase rating up to 1600A and it can be supplied with ACB,MCCB or fuse protection.

3.2.1 Wiring installation Feeder Pillar.

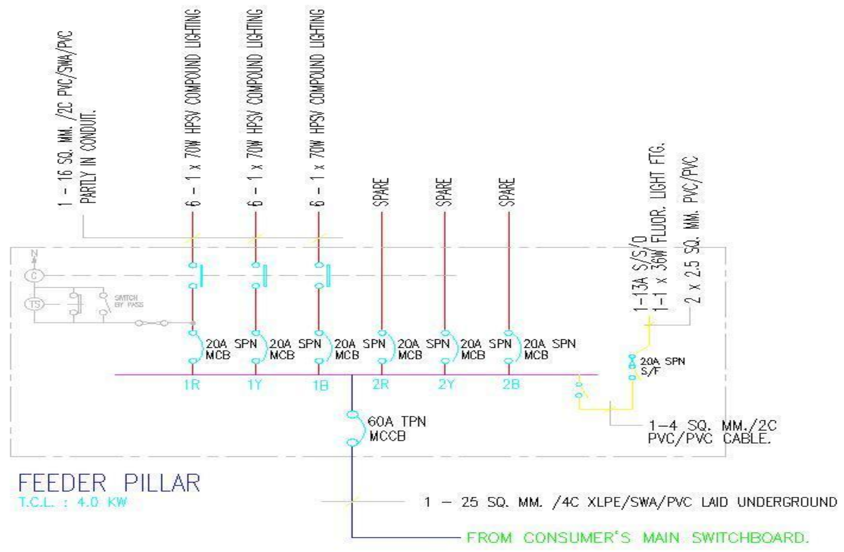


Figure 1.2: Compound Lighting Feeder Pillar Schematic Image

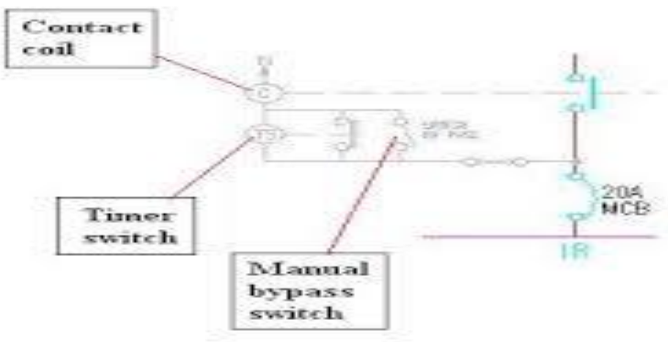


Figure 1.3: Timer schematic image.

3.2.2 The time switch.

-Common to all street lighting and compound lighting.

-To eliminate the need for human actions to turn ON and OFF the lighting at the intended operation hours.

3.2.3 Switch bypass @ manual bypass switch.

-Is used for maintenance or trouble-shooting works.

-It simply bypass the time switch to turn ON or OFF the external lighting or street lighting.

3.2.4 The contact coil

-The load currents to the lighting cables are usually higher than the current that the timer contacts can handle.

-Another coil with heavy current contacts needs to provided to switch the heavy currents.

3.2.5 Safety step before troubleshooting:

i) No material or equipment that the person is holding,carrying or is supported by,which is not insulated for the voltage concerned.

ii)Know how the circuit is function.

3.2.6 Troubleshooting

i) Look for the trip at feeder pillar.

ii)Turn OFF all circuit breaker and set the time at time switch.

iii)Turn ON from main switch and turn ON one by one the circuit breaker with bypass at timer to ON the light.

iv)If found the trip at circuit breaker,that's mean one of the lamp have the problem.

v)Lastly, turn OFF the bypass at time switch.

3.3 Substation

Figure 1.4 : substation at politeknik banting

Definition

A substation is a part of an electrical generation, transmission, and distribution system. Substations transformvoltage from high to low, or the reverse, or perform any of several other important functions. Between the generating station and consumer, electric power may flow through several substations at different voltage levels. A substation may include transformers to change voltage levels between high transmission voltages and lower distribution voltages, or at the interconnection of two different transmission voltages.

Substations may be owned and operated by an electrical utility, or may be owned by a large industrial or commercial customer. Generally substations are unattended, relying on SCADA for remote supervision and control.

The word substation comes from the days before the distribution system became a grid. As central generation stations became larger, smaller generating plants were converted to distribution stations, receiving their energy supply from a larger plant instead of using their own generators. The first substations were connected to only one power station, where the generators were housed, and were subsidiaries of that power station

3.4 Component inside Substation

3.4.1Transformer.

Figure 1.5: Transformer inside TNB substation.

This transformer can steps down the high voltage electricity to the low voltage electricity

3.4.2 Vacuum circuit breaker.

Figure 1.6: Vacuum circuit breaker.

3.4.3 Power factor correction board.

Figure 1.7: Power factor correction board.

Power Factor Definition:

Power factor is the ratio between the KW and the KVA drawn by an electrical load where the KW is the actual load power and the KVA is the apparent load power. It is a measure of how effectively the current is being converted into useful work output and more particularly is a good indicator of the effect of the load current on the efficiency of the supply system.

Power Factor Improving

1.Check if required kVAr of capacitors are installed.

2.Check the type of capacitor installed is suitable for application or the capacitor are derated.

3.Check if the capacitors are permanently ‘ON’. The Capacitor are not switched off.

4.When the load is not working, under such condition the average power factor is found to be lower side.

5.Check whether all the capacitors are operated in APFC depending upon the load operation.

6.Check whether the APFC installed in the installation is working or not.

7.Check if the Load demand in the system is increased.

8.Check if power transformer compensation is provided.

3.4.4 Earth system.

Figure 1.8: Earthing system inside the substation.

Definition

Earthing system connects specific parts of that installation with the Earth’s conductive surface for safety and functional purposes.

Function of Earthing system

-To protect a single-wire earth return power and signal lines, such as were used for low wattage power delivery.

-As ancillary voltage balance for other kinds of radio antennas, such as dipoles.

-As the feed-point of a ground dipole antenna for VLF and ELF radio.

3.5 Fire protection system

Politeknik banting is equipped with hose reel system, fire alarm system,heat and smoke detectors,portable fire extinguishers and other passive fire systems.The walls are protected with Fire Rated Walls to ensure the building is able to withstand the fire for a certain of time to allow occupants to escape.

Every fire exit and escape staircase is equipped with a “Keluar” sign to guide the building occupants to the exit path. Besides, a simplified floor plan with indications on the emergency exit is located next to the lift on every level of the building. Emergency light are also installed at each areas of the building to ensure illumination for the building occupants to the nearest exits in thick smokes or sudden blackout.

3.5.1 Active Fire Protection

An action is required for active fire protection systems to work,either it to be by manual,electrical or mechanical. Active system detects fire through detector that will send signals to devices sush as alarm bell to alert the building occupants. This system then controls fire by activating fire shutter doors to limit the spread of fire and smoke to other area of the building. It suppresses or extinguish fire through carbon dioxide system, sprinklers, hose reel system, riser system and the use of fire hydrant.

3.5.2 Passive Fire Protection

Passive system does not require external power or any activation and can be grouped into three categories according to its purposes:

I) Limiting the growth rate of fire

Ii) Compartmentation of fire

Iii) Providing emergency escape from fire areas

Passive system slows down fire with fire-resistant walls, floors, doors or spray-on fireproofing mixture on critical members such as beams and columns. This protects the building from collapsing due to the weaken building parts in high temperature condition and provides building occupants with more time to evacuate.

3.5.3 Active Fire Protection

3.5.4 Hose Reel System.

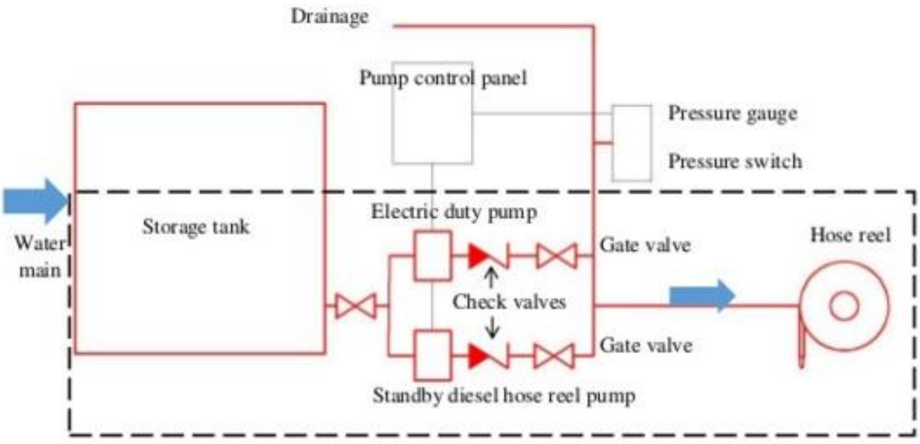


Figure 1.9 : schematic drawing from hose reel system

3.5.5 Pumps.

Figure 2: Electric duty pump

-Can automatic OFF when the water have reach limit to pump.

-Lost cost to operate.

Figure 2.1:Standby diesel hose reel pump

-Manual OFF.

-Using too much cost to operate.

3.5.6 Pressure gauge and pressure switches.

Figure 2.2: Pressure gauge and pressure switches

Figure 2 is the adjusted pressure field setting of the pressure switch. This triggers the automatic operation of the pump to supply water for the hose. On the other hand,the pump will shut OFF when the pressure is higher than the pre-set range.

Figure 2.1: The control panel is to operate the pumps by manual

and to check the failure either pump.

3.5.7 Hose Reel

Figure 2.2: Hose reel

-Wet riser,dry riser,sprinkler and other fire installation pipes and fittings shall be painted red.

-All cabinets and areas of recessed in walls for location of fire installations and extinguishers shall be clearly identified to the satisfaction of the Fire Authority or otherwise clearly identified.

3.6 Fire Alarm system

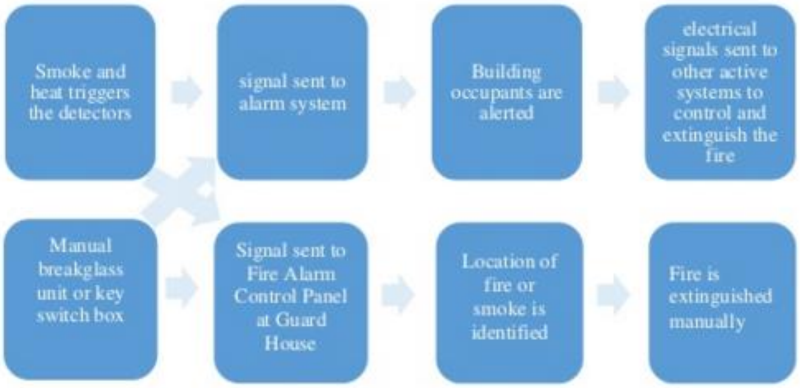


Figure 2.4:Flow for fire alarm system.

3.6.1 Manual Break glass Unit and Key Switch Box

Figure 2.5: Break glass unit

The Fire alarm system are alerts the building occupants when there is a fire outbreak so that immediate safety measure and fire fighting action can be taken. This system can be operated either automatically through the detectors or manually by breaking the glass of the break glass unit to activate the fire alarm.

Figure 2.6: Manual key switch boxes located outside of TNB switch room and transformer room.

Individual manual key switch boxes are installed outside of each switch room and transformer room. This enables quick activation of fire alarm and release of carbon dioxide gas when the room is on fire by turning the key of the key switch box. Key switch boxes are used here instead of manual breakglass unit is to prevent anyone to intentionally activate the carbon dioxide system without emergency as only permissioned individuals and the Fire Authority have the key.

3.6.2 Portable fire extinguisher

Figure 2.7: Portable fire extinguisher

Portable extinguisher shall be provided in accordance with the relevant codes of practice and shall be sited in prominent positions on exit routes to be visible from all directions and similar extinguishers in a building shall be of the same method of operation.

3.7 Panel board

Figure 2.8: Distribution board/panel board

A distribution board can be known as panel board,breaker panel or electric panel.It is a component of an electricity supply system that divides an electrical power feed into subsidiary circuits, while providing a protective fuse or circuit breaker for each circuit in a common enclosure.

3.8 Component inside panel board.

Figure2.9: Circuit breaker (CB).

Definition

A circuit breaker is an automatically operated electrical switch designed to protect an electrical circuit from damage caused by excess current, typically resulting from an overload or short circuit. Its basic function is to interrupt current flow after a fault is detected. Unlike a fuse, which operates once and then must be replaced, a circuit breaker can be reset (either manually or automatically) to resume normal operation. Circuit breakers are made in varying sizes, from small devices that protect low-current circuits or individual household appliance, up to large switchgear designed to protect high voltage circuits feeding an entire city. The generic function of a circuit breaker, RCD or a fuse, as an automatic means of removing power from a faulty system is often abbreviated as ADS (Automatic Disconnection of Supply).

3.8.1 Type of circuit breaker

MCB(Miniature circuit breaker).

Figure 3 : Miniature circuit breaker.

-Trip normally not adjustable.

-Rated current not more than 100A.

-Thermal or thermal-magnetic operation.

-Low energy requirement like home wiring or small electronic circuit.

MCCB(Moulded Case Circuit Breaker).

Figure 3.1: Moulded case circuit breaker.

-Rated current up to 1000A.

-Trip current may be adjustable.

-Providing energy for high-power equipment.

ELCB(Earth Leakage Circuit Breaker).

Figure 3.2: Earth leakage circuit breaker.

-ELCB is working based on Earth Leakage current.

RCCB/RCD(Residual Current Circuit Breaker).

Figure 3.3:Residual current circuit breaker.

-It trip the circuit when there is earth fault current.

RCBO(Residual Circuit Breaker with Overload).

Figure 3.4: Residual circuit breaker with overload.

-To protect Life wire.

-To protect an overload on a circuit-overcurrent and short circuit.

3.8.2 Vacuum CB(Vacuum circuit breaker).

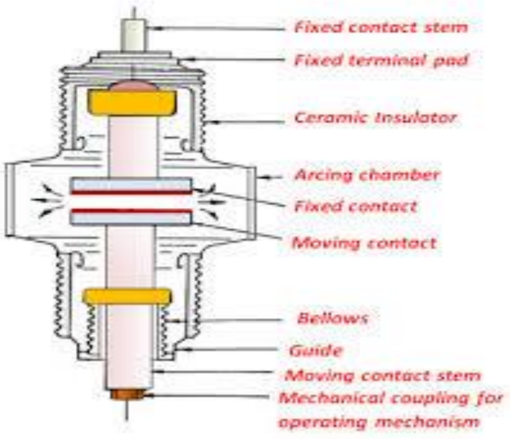


Figure 3.5: Vacuum circuit breaker.

Figure 3.6: Vacuum CB construction.

-A breaker which used vacuum as an arc extinction medium.

-The fixed and moving contact is enclosed in a permanently sealed vacuum interrupter.

-Medium voltage range from 11kV to 33kV.

It has two phenomenal properties:

1.High insulating strength.

-In comparison to various other insulating media used in VCB is a superior dielectric medium.

2.When an arc is opened by moving apart the contacts in a vacuum.

3.8.3 Multi Connector/brass bar.

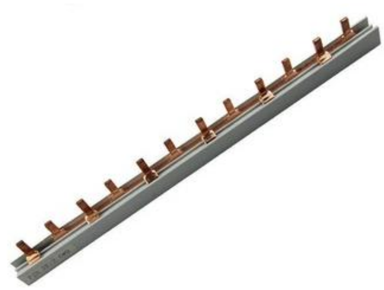


Figure 3.7: brass bar

3.8.4 Relays.

Figure 3.8: relays

Definition Relays are switches that open and close circuits electromechanically or electronically.

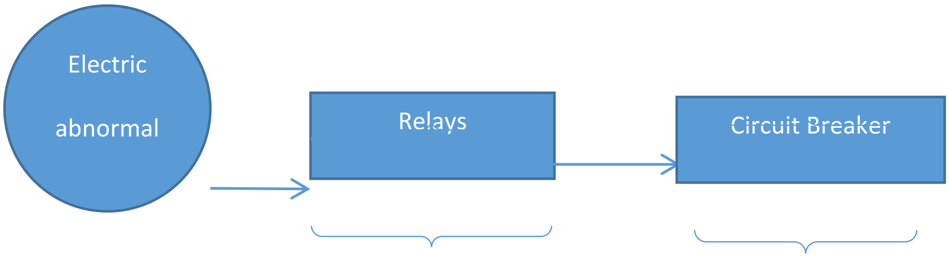
Have 2 types:-

-Normally Open (NO).

-Normally Close (NC).

Protective relays can prevent equipment damage by detecting electrical abnormalities, including overcurrent, undercurrent, overloads and reverse currents.

3.8.5 Relays function



Detect the abnormal in

Electric and make the circuit

Breaker to open the circuit

Circuit breaker will open the

circuit.That's mean when

circuit breaker open the circuit,
It will not conduct electricity and

Stopped the flow of current in the

loop.

3.8.6 Voltage in Malaysia.

Single phase - 240V

3 phase - 415V

Frequency-50Hz

Basic for comparison	Single phase	3 phase
Definition	The power supply through one conductor.	The power supply through 3 conductor.
Number of wire	Require two wires for L(life),N(neutral) to completing the circuit.	Requires four wires for R(red),Y(yellow),B(blue), and N(neutral) to completing the circuit.
Voltage	Carry 240V	Carry 415V
Phase name	Split phase	No other name
Power transfer capability	Minimum	Maximum
Network	Simple	Complicated
Power failure	Occurs	Do not occurs
Loss	Maximum	Minimum
Efficiency	Less	High
Economical uses	Less for home appliances	More in large industries and for running heavy loads.

Table 1.8:Comparison between single phase and 3 phase.

3.8.7 Problems during industrial training.

-Our panel board for air condition in Hanger have been entered by water because the water come from the leaking gutter. All the circuit breaker has been moistened with water, and the short circuit occur in there. Small blast happen when turn on the main circuit breaker, it happen because water at circuit breaker has make the electric from any wire to combine together.

Figure 4:The problem panel board.

Figure 4:Damaged multi connector.

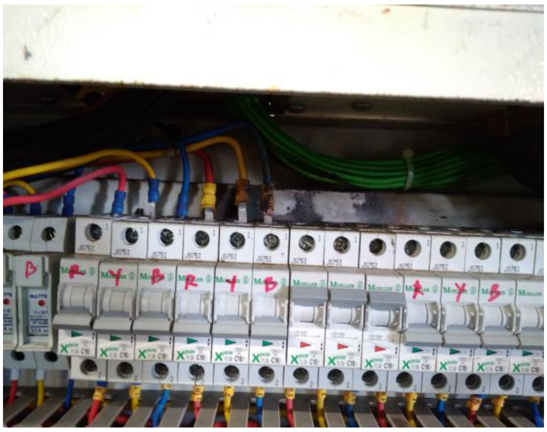


Figure 4.1:Damaged area in circuit breaker.

Figure 4.2:Damaged circuit breaker.

3.8.8 Safety step before troubleshooting.

i) No material or equipment that the person is holding,carrying or is supported by,which is not insulated for the voltage concerned.

3.8.9 Troubleshooting.

i)Make sure the main electric for the damaged panel board has been turned off at substation TNB and turned off all circuit breaker in panel board.

ii)Always use the test pen to check the supply at main circuit breaker.

iii)Open the main circuit breaker and elcb circuit breaker and the damaged component.

iv)Use the blow gun air compressor to blow the wet circuit breaker until it dry and blow component inside panel board until dry.

v)Fix the leaking gutter and install back the circuit breaker to the palace of origin.

vi)Install the damaged component with new one.

vii)Turned on the main supply for panel board from substation TNB, turned on all circuit breaker and test the supply at main circuit breaker.

3.9 Telephone plug

Definition

A telephone plug is a type of connector used to connect a telephone set to the telephone wiring inside a building, establishing a connection to a telephone network. It is inserted into its counterpart, a telephone jack, commonly affixed to a wall or baseboard.

3.9.1 Component of telephone plug

Modular phone connector

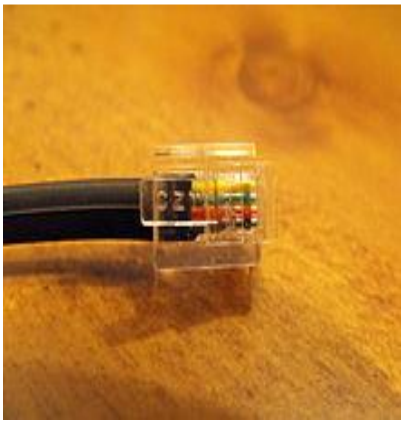


Figure4.3: Modular phone connector

Socket

Figure4.4:Socket

Telephone junction box

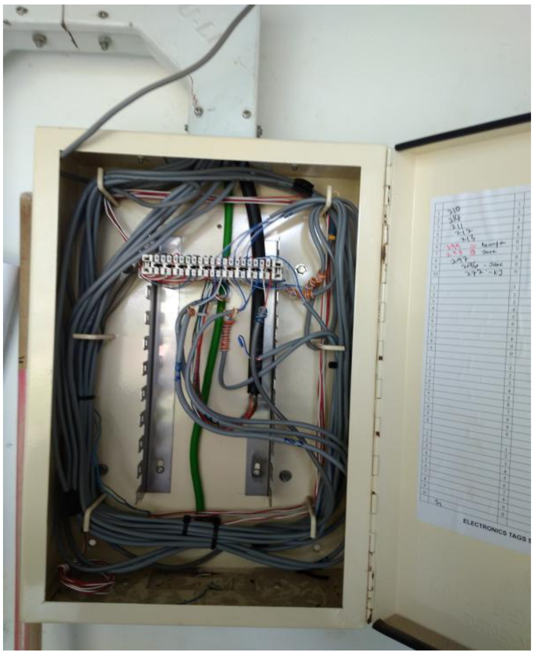


Figure 4.5:Telephone junction box

Definition

A junction box is a connection point used in supplying telephone service to various points in a building. Providing an interface between exterior and interior wiring, junction boxes contain terminals for each of the four colored phone wires and allow for expansion of a phone system.

3.9.2 Tool & equipment for making a new line telephone.

Crimping tool

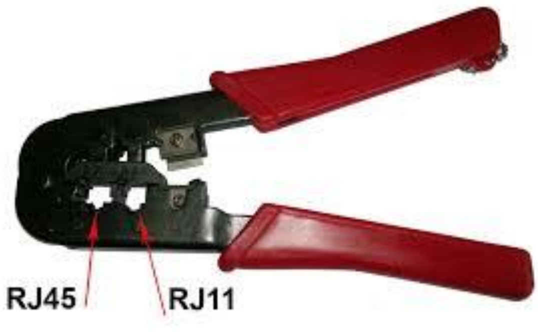


Figure4.6:Crimping tool

-It have a two hole (RJ45 & RJ11) to easier for crimp with wire.

Multimeter

Figure 4.7: Multimeter

-To check the same value voltage at telephone juction box for connect with the correctly line.

3.10 Air compressor

Definition

Air compressor is a device that converts power(using an electric motor, diesel or gasoline engine, etc.)into potential energy stored in pressurized air(compressed air). An air compressor forces more and more air into a storage tank, increasing the pressure. When tank pressure reaches its upper limit the air compressor shuts off. The compressed air,then,is held in the tank until called it to use. When tank pressure reaches its lower limit, the air compressor turns on again and re-pressurize the tank.

3.10.1 Part of air compressor

Check valve **Compressor pump**

Figure 4.8: Silver check valve figure4.9: compressor pump

Pump flywheel **Air filter**

Figure 5: pump flywheel

Figure 5.1: air filter

Motor

Figure 5.2:motor

Pressure switch

Figure 5.3:pressure switch

Gauge

Tank

Figure 5.4: gauge

Figur 5.5 : tank

3.10.2 Difference between 2 stroke and 4 stroke.

Four stroke engine	Two stroke engine
-It has one power stroke for every two revolutions of the crankshaft	-It has one power stroke for each revolution of the crankshaft
-Engine is heavy	-Engine is light

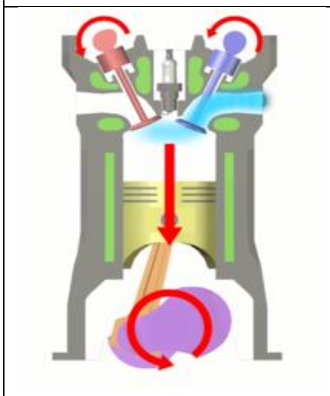
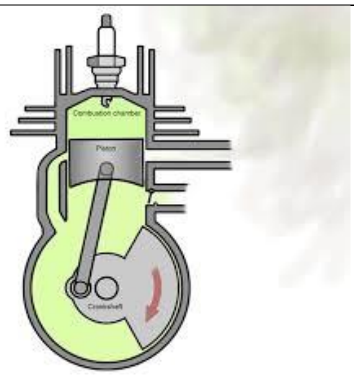
-Engine design is complicated due to valve mechanism.	-Engine design is simple due to absence of valve mechanism.
-More cost	-Less cost
-Less mechanical efficiency due to more friction on many parts.	-More mechanical efficiency due to less friction on few parts.
-Less noise is created by engine.	-More noise is created by engine.
-Consumes less lubricating oil.	-Consumes more lubricating oil.
-Engine runs cooler.	-Engine runs hotter.
-Engine is water cooler	-Engine is air cooled
	

Table 1.9: Difference 2 stroke and 4 stroke.

-Air compressor use Four stroke engine application to compress the air and use the AC motor to move the rotor.

3.10.3 Problems during industrial training.

Air compressor in store have been damaged because it run non stop. It cause the capacitor damage, it run non stop because one of the gasket has loose and the gauge can’t reach the limit to turn off the air compressor.

Figure 5.6:Damage capacitor.

Figure 5.7:Compressor in store

Figure 5.8 :New capacitor

3.10.4 Troubleshooting

I)Turn off the supply at socket.

Ii)Open and change the damage capacitor with new one.

Iii)Change the pressure switch with new one and set the pressure limit.

v)Turn on the supply and test the compressor if it can stop at the limit set of the pressure.

CHAPTER 4

4.1 Conclusion and recommendation

Industrial training is very much essential for Electrical Students. It is also a great opportunity to acquire practical knowledge. During my training period, in the industry I acquired lots of experiences in Politeknik Banting Selangor. This will help me to clarify my theory knowledge. I hope and pray this experience during industrial training will help me much in my future profession.

During our training period, we had seen the various instruments and apparatus in the industry. The important thing that instruments usually use for troubleshooting the electrical problems is test pen and multimeter. Our development and maintenance unit will analysis the complaint letter and go to the location for maintenance the damage items.

This training have taught me to learn more than 1 basic course other than electrical course such as mechanical or civil because in real job, we will be instructed by employer to do work other than our course to look at skills in various jobs. My supervisor is technician in mechanical engineering but he learn all about electrical and civil to make sure more understanding when dealing with contractor.

I had been monitored by supervisor for safety in doing many jobs. Safety first always will be our first rule before doing anything maintenance to make sure the job can smoothly done. Sometimes the job was difficult to finish it and the technician will make a report to buy a contractor to solve the problems.

Overall, This development and maintenance unit need more worker to handle the difficult job for give more efficiency. Need a worker in various field that can help other worker to know better in troubleshooting and gained more knowledge. This unit need a lot of budget for maintenance job to give a good condition in surrounding Politeknik Banting Selangor.

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